

Figure 5: Colours and markers in plots

This line plots a sine wave at a position $a=0$ with the marker 'o' in red colour. The remaining five lines also plot sine waves with different markers and different colours. There are other markers also available for use in Matplotlib like ^, <, +, etc. On executing the program, the output is the image shown in Figure 5.

Saving Matplotlib figures

Now consider the program `plot5.py` shown below. It will save the figure generated by Matplotlib into a specific output format. The output formats supported by Matplotlib include eps, jpeg, jpg, pdf, pgf, png, ps, raw, rgba, svg, tif, tiff, etc. If you observe this list carefully, you will see that the output formats of Matplotlib include both raster graphics and vector graphics formats. Do remember that vector graphics clearly outperform raster graphics as far as image quality is considered, especially when enlarged to large sizes. Apart from Matplotlib, PGF/TikZ and PSTricks can also be used to create vector graphics. So, mastering any one of these tools will help you create good quality images, which is almost always a necessity while publishing journals and magazines. The program `plot5.py` also illustrates certain other necessary features of Matplotlib like setting plot labels and the plot title.

```
import matplotlib.pyplot as plt
import numpy as np

a = np.linspace(-10,10,1000)
fig = plt.figure()
plt.title("Sine Wave")
plt.xlabel("a")
plt.ylabel("Sine(a)")
plt.plot(a,np.sin(a),color='red')
fig.savefig('Image.png')
```

The line of code:

```
fig = plt.figure( )
```

...creates a new figure. The line of code:

```
fig.savefig('Image.png')
```

...saves the plot as *Image.png*. The line of code:

```
plt.title("Sine Wave")
```

...sets the title of the image as *Sine Wave*. The lines:

```
plt.xlabel("a")
```

...and:

```
plt.ylabel("Sine(a)")
```

...set labels on the x-axis and y-axis. On executing the program, `plot5.py` will not display any image on the screen, but will generate a file named *Image.png* on the same directory in which the program `plot5.py` is stored and executed. Figure 6 shows the image *Image.png*.

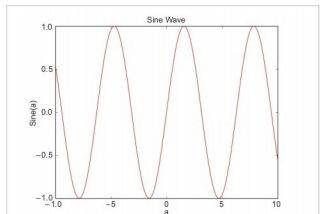


Figure 6: Image.png

Histograms and scatter plots in Matplotlib

Consider the program `plot6.py` shown below, which will draw a histogram and a scatter plot using Matplotlib:

```
import matplotlib.pyplot as plt
import numpy as np
a = np.random.randn(1000)
b = np.random.randn(100)
c = np.random.randn(100)
plt.subplot(2,1,1)
plt.hist(a,color='green')
plt.subplot(2,1,2)
plt.scatter(b,c,marker='o',color='red')
```